

CSCI 164: Machine Learning Report

Raheem Saeed | Harman Kandola | Hekmatullah Najm

I analyzed the **Turkish Crowdfunding Startups** dataset (UCI ID 1025), which comprises 1,628 campaigns and **38** features spanning categorical platform details, project metrics, monetary targets, and free-text descriptions. After imputing a single missing region value, dropping raw date fields, and engineering three new features (log-transformed target and raised amounts plus numeric percent funded), we constructed a unified scikit-learn **ColumnTransformer** to one-hot encode categorical variables and standard-scale numeric ones.

Using this pipeline, we trained and tuned three classifiers via 5-fold **GridSearchCV** (F1 scoring):

- **Logistic Regression** ($C=10$)
- **k-Nearest Neighbors** ($k=9$, uniform weights)
- **Histogram-based Gradient Boosting** (learning_rate=0.1, max_depth=3, L2 regularization=0)

On a 30 % held-out test set, **logistic regression** emerged as the top performer with **F1 = 0.991** and **ROC-AUC = 0.997**, misclassifying only 3 out of 489 samples. **HistGradientBoosting** was closely followed (F1 = 0.987, AUC = 0.995), while **k-NN** trailed (F1 = 0.878, AUC = 0.970), likely due to the high-dimensional one-hot feature space.

Our results represent a substantial improvement over prior work by Kilinc & Aydin (2023), who reported **82.6 %** accuracy using **k-NN** on an earlier snapshot of the same data. We attribute this gain to targeted feature engineering (e.g., log-transforms), a dense encoding strategy, and rigorous hyperparameter tuning. These findings demonstrate that when paired with thoughtful preprocessing, simple, interpretable models can deliver near-perfect predictive power, offering crowdfunding platforms and entrepreneurs a reliable tool for campaign risk assessment and strategic planning.